Aditya Das Gupta

Professional Profile

Electrical and Electronic Engineering graduate with strong technical expertise in embedded systems, IoT development, and software engineering. Proven ability to design, implement, and manage engineering projects with excellent problem-solving and analytical skills.

Contact

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Technical Skills

- **Programming:** Python, C++
- **Embedded Systems:** Arduino, ESP8266, Microcontrollers
- **Circuit Design:** Proteus, LTSpice, Tanner EDA
- **Simulation Tools:** MATLAB, Microwind, DSCH
- **Web Development:** HTML, CSS, WordPress, MQTT
- **Mobile Development:** Android, Kotlin

Engineering Skills

- IoT System Design
- PCB Design & Layout
- Sensor Integration
- Signal Processing
- Control Systems
- Data Analysis

Professional Skills

- Project Management
- Technical Documentation
- Team Leadership
- Problem Solving
- Research & Development
- Quality Assurance

Certifications

- Industrial Technology Training (TICI)
- IEEE Conference Presenter

Languages

Bangla: Native
English: Fluent

Education

B.Sc. in Electrical & Electronic Engineering — **Feb 2020 – Nov 2025**
Barisal Engineering College, University of Dhaka Technology Unit
GPA: 3.26/4.00

Higher Secondary Certificate — **2019**
Chattogram Cantonment Public College (CCPC)
GPA: 4.92/5.00

Secondary School Certificate — **2017**
Begum Gul Chemonara Academy School & College (BGCA)
GPA: 5.00/5.00

Engineering Projects

Smart IoT-Based Water Tank Monitoring and Alert System — **2024**
◇ Designed ESP8266-based water level monitoring with Android app integration
◇ Implemented MQTT communication for real-time monitoring and alerts

IoT-Based Productivity Monitoring and Management System — **2024**
◇ Developed ESP8266 device with Android app for productivity tracking
◇ Integrated GSM module for SMS control and local data storage

Automated IoT Plant Health Monitoring System — **2023**
◇ Designed cactus monitoring system with automated watering and sensor tracking
◇ Integrated Blynk and Google Sheets for remote monitoring and data logging

Research Experience

Undergraduate Thesis Researcher — **Sep 2024 – Aug 2025**
Barisal Engineering College
◇ Developed Explainable AI Framework for Non-Invasive Blood Glucose Prediction
◇ Applied machine learning techniques and statistical analysis for healthcare applications

Professional Experience

Industrial Training Participant — **May 2025 (2 Weeks)**
Training Institute for Chemical Industries, Narsingdi
◇ Gained hands-on experience with industrial electrical systems and instrumentation
◇ Learned control processes and industrial automation systems

Publications

Feature-Optimized Blood Glucose Level Prediction with XAI — **2025**
4th IEEE International Conference on Signal Processing, Information, Communication and Systems
DOI: 10.1109/SPICSCON69221.2025.11504022

Development and Validation of ML Model for Non-Invasive Hypoglycemia Detection — **2025**
28th IEEE International Conference on Computer and Information Technology
DOI: 10.1109/ICCIT68739.2025.11491420

Interpretable ML Framework for Patient Risk Prediction using XAI — **2025**
4th IEEE International Conference on Robotics, Automation, AI and IoT
DOI: 10.1109/RAAICON69033.2025.11502480

Modeling and Controlling Lead-Acid Battery Storage in a Hybrid Microgrid **2025**
IEEE International Conference on Power, Electrical, Electronics & Industrial Applications (PEEIACON 2025) *(Accepted)*

Leadership & Activities

General Secretary — **Jul 2024 – Aug 2025**
BEC Film & Photography Society

Editor — **May 2024 – Present**
Rotaract Club of Chittagong, Karnaphuli

Executive Member — **2024 – 2025**
BEC Electronics Club, Research and Innovation Club, Sports Club, Cultural Club

Awards & Honors

Compensatory Scholarship — **2020 – 2025**
Barisal Engineering College for Academic Performance

Honorary Award — **2017**
Cambridge School and College for Outstanding SSC Results